

Arka Ghosh

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CAREER OBJECTIVE

Machine Learning Engineer with 2+ years of experience developing end-to-end ML solutions. Experienced in Computer Vision, Deep Learning, and GenAI, with a focus on building practical and scalable AI systems.

EDUCATION

•FUTURE INSTITUTE OF TECHNOLOGY MAKAUT

B.TECH IN COMPUTER SCIENCE AND ENGINEERING (AI & ML)
JUL 2024 | Kolkata, West Bengal
CGPA: 8.81

•NAVA NALANDA HIGH SCHOOL,WEST BENGAL/WBCHSE

March 2020 | Kolkata, West Bengal
12th: 83.33%

•NAVA NALANDA HIGH SCHOOL,WEST BENGAL/WBBSE

March 2018 | Kolkata, West Bengal
10th: 74.14%

LINKS

🐙 Github:// [arkaghosh566](#)
🌐 LinkedIn:// [ARKA GHOSH](#)
🌐 Portfolio:// [Arka Ghosh](#)

LANGUAGES

- English: Full Professional Proficiency
- Bengali: Native or Bilingual Proficiency
- Hindi: Limited Working Proficiency

SKILLS

PROGRAMMING

Languages:

• Python • C++ • Java • JavaScript • SQL

Libraries & Frameworks:

• NumPy • Pandas • Scikit-learn • TensorFlow • PyTorch • OpenCV • Keras
• Matplotlib • Seaborn

Databases:

• SQL • MongoDB • SQLite

Deployment Tools:

• Flask • Docker • Streamlit

WORK EXPERIENCE

CAPSITECH | Assistant ML Engineer

June 2024 – April 2026 | Jodhpur, Rajasthan

- Developed a real-time computer vision employee tracking system using SCRFD for face detection and ResNet50-based ArcFace for long-distance identity recognition.
- Built a document QA assistant using FAISS-based semantic retrieval and LLM inference with strict prompt constraints to reduce hallucinations.
- Built a YOLO-based hair follicle detection system on a custom dataset, followed by Hough Line Transform analysis for dermatological applications.
- Developed a crowded-environment person counting system using ResNet50 with Feature Pyramid Network (FPN) and real-time inference optimization.
- Extracted gait silhouettes using YOLOv8 and Segment Anything Model (SAM) for downstream feature extraction.
- Researched and implemented SORT, DeepSORT, ByteTrack, BoT-SORT, and OC-SORT for multi-object tracking.

PROCENTRIS INDIA PVT. LTD. | Software Engineer (AIML)

May 2026 – Present | Kolkata, West Bengal

- **OmniTranslate:** Built an end-to-end document translation pipeline for PNG/JPG/PDF inputs using OCR and Gemini Flash, reconstructing translated PDFs while preserving original layout, formatting, and text positioning.
- **InvoiceAI:** Developed an AI-powered medical invoice and insurance-claim pipeline that extracts data from PDF/image inputs and maps it to a predefined JSON schema using Gemini Flash, with UI-based validation of beneficiary, amount, and claim details.
- **EmsAI:** Built an incremental email-thread summarization system that updates stored summaries using previous summaries and newly received conversations; integrated Gemini and Qwen through a switchable LLM architecture.
- **Document Authenticity Analysis:** Built a fraud-analysis pipeline for image, PDF, DOC, and DOCX files using ExifTool and supporting libraries to extract metadata and content, followed by Gemini-based analysis of inconsistencies and anomalies to generate fraud prediction and confidence scores.

CERTIFICATIONS

MACHINE LEARNING

Oct 2022 | COURSERA

- Decision Trees, Neural Networks, Logistic Regression, Recommender Systems

CONVOLUTIONAL NEURAL NETWORKS

Jul 2023 | COURSERA

- Deep Learning, Facial Recognition, CNN, TensorFlow, Object Detection

GOOGLE DATA ANALYTICS

Sept 2023 | COURSERA

- Data cleaning, analysis, visualization with SQL, R, Tableau

GOOGLE ADVANCED DATA ANALYTICS

Sept 2023 | COURSERA

- Regression, ML models, statistical methods, data visualizations